

2012

6 4 24

1 $4 \sqrt{4}$

A 2

B ± 2

C 2

D $\pm \sqrt{2}$

2 4

A $a+a^2$

B $2a^3-6a^3$

C $a-1^2-a^2-1$

D $a^3 \div a-a^2$

3 4 Rt△ABC ∠C=90° ∠A=∠B=∠C $a:b:c=a^3:b^4$ ∠B

A $\frac{3}{5}$

B $\frac{4}{5}$

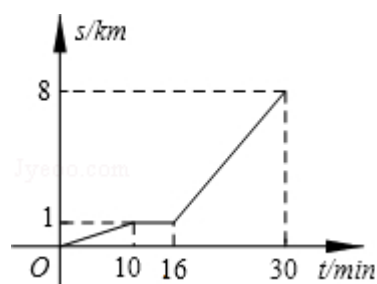
C $\frac{4}{3}$

D $\frac{3}{4}$

4 4

s km

t min



A 8km 30min

B 6min

C 100m/min

D 350m/min

5 4 $x^2-x+2=\frac{1}{x^2-x}$ $y=x^2-x$ y

A $y^2-2y+1=0$

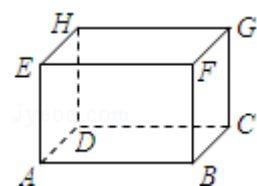
B $y^2-2y+1=0$

C $y^2+2y+1=0$

D $y^2+2y+1=0$

6 4 ABCD EFGH

AB



A EA

B GH

C HC

D EF

□□□□□□□□□□ 12 □□□□ 4 □□□□ 48 □□

7 4 $y=\frac{\sqrt{x+2}}{x-1}$ x _____

8 4 2010 11 1370536875

9 4 $\frac{2}{x-1}=1$ _____

10 4 x^2-2x+1 _____

11 4 $x^2-4x+a=0$ a _____

12 4 $y=\frac{m-3}{x}$ x 0 y x m _____

13 4 “ ” “ ” x $60\leq x < 100$

$60\leq x < 70$	30	0.15
$70\leq x < 80$	m	0.45
$80\leq x < 90$	60	n
$90\leq x < 100$	20	0.1

n _____

14 4 6 120 180

$x \div$ _____

15 4 $ABCD$ $AB\parallel CD$ $E F$ $AD \parallel BC$ $\overrightarrow{AB}=\overrightarrow{a}$ $\overrightarrow{CD}=\overrightarrow{b}$ $\overrightarrow{a}$ $\overrightarrow{b}$
 $\overrightarrow{EF}$ _____

16 4 $R=r$ $x^2-5x+6=0$ 1 _____

17 4 $\triangle ABC$ G $\triangle ABC$ G $DE\parallel BC$ AB D AC E
 $\triangle ABC$ 9 $\triangle ADE$ _____

18 4 $ABCD$ $AD=4$ $CD=2$ $AD \perp A$ D $CB \perp P$ $\angle DPC$ _____

□□□□□□□□□□ 7 □□□□ 78 □□

19 10 $(-\frac{1}{4})^{-1}+\sqrt{27}+|\sqrt{3}-3|+\cot 30^{\circ}$

20 10
$$\begin{cases} 3x-7<2(1-3x) \\ \frac{x-3}{2}+1\leqslant \frac{3x-1}{4} \end{cases}$$

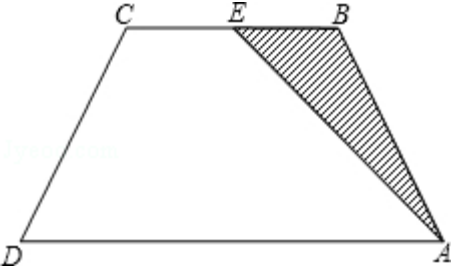
21 10 $87m$ $AD\parallel BC$

$10m$ AB $i=\frac{5}{3}$ AB

AE $i=\frac{5}{6}$

1 AB

2 EC $2.7m$ AD

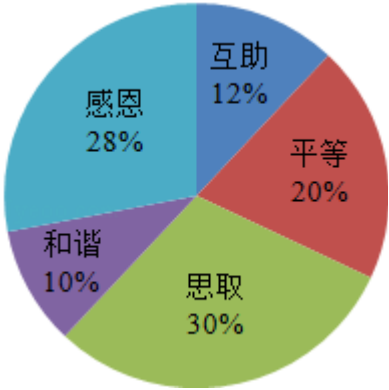


22 10 “ ” 2 5 50

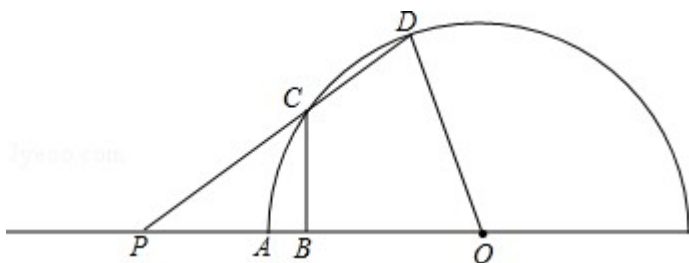
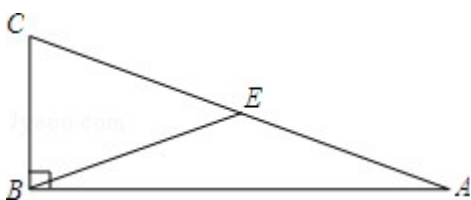
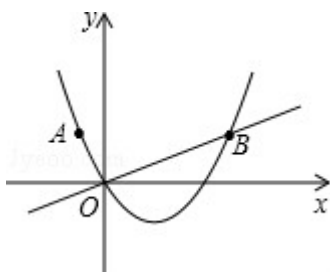
1 “ ” _____ “ ” _____

2 1500 “ ” _____

3 5 “ ” “ ”



23 12 A 1 2 O $l\perp OA$ A O B



2012 □ □ □ □ □ □ □ □ □ □ □ □ □ □ □

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□ □ □ □ □ □ □ □ □ □ 6 □ □ □ □ 4 □ □ □ □ 24 □ □

1 $4 \sqrt{4}$

A 2

B ± 2

C 2

D $\pm \sqrt{2}$

22

$\sqrt{4} 4$

$\sqrt{4} 2$

A

2 4

A $a+a^2$

B $2a^3-6a^3$

C a^2-1-a^2-1

D $a^3 \div a^2$

35

47

48

$4C$

A $a+a-2a \neq a^2$

B $2a^3-8a^3 \neq 6a^3$

C $a^2-1-a^2-2a+1 \neq a^2-1$

D $a^3 \div a^2$

D

3 4 Rt△ABC $\angle C = 90^\circ$ $\angle A = \angle B = \angle C$

$a^3-b^3 = (a-b)(a^2+ab+b^2)$ $\angle B$

A $\frac{3}{5}$

B $\frac{4}{5}$

C $\frac{4}{3}$

D $\frac{3}{4}$

KQ

T1

c

Rt△ABC $\angle C = 90^\circ$ a^3-b^4

$\therefore c = 5$

$$\therefore \sin B = \frac{4}{5}$$

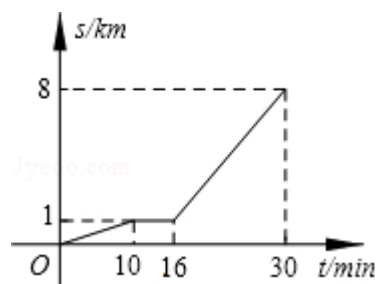
B

C

4 4

s km

t min



A 8km 30min

B 6min

C 100m/min

D 350m/min

E6

16 31

8km

A 8km 30min A

B 10min 16min 6min B

C 10min 1000m 100m/min C

D 30 16 min 8 1 km 7000÷14 500m/min D

D

$$5 \quad 4 \quad x^2 - x + 2 = \frac{1}{x^2 - x} \quad y \quad x^2 \quad x \quad y$$

A $y^2 - 2y - 1 = 0$ B $y^2 - 2y + 1 = 0$ C $y^2 + 2y + 1 = 0$ D $y^2 + 2y - 1 = 0$

B4

$$y \quad x^2 \quad x \quad y + 2 \quad \frac{1}{y}$$

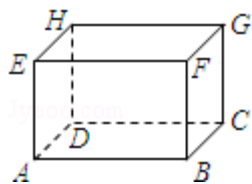
$$y+2\frac{1}{y}$$

$$y^2+2y-1$$

$$y^2+2y-1=0$$

D

6. 4. $ABCD \cong EFGH$ AB



A. EA

B. GH

C. HC

D. EF

I1

J3

J7

1

A. EA

AB

B. GH

AB

C. HC

AB

D. EF

AB

C

□□□□□□□□□□ 12 □□□□ 4 □□□□ 48 □□

$$7-4 \quad y=\frac{\sqrt{x+2}}{x-1} \quad x \quad \underline{x \geq 2 \quad x \neq 1}$$

E4

0 0

$$\begin{cases} x+2 \geq 0 \\ x-1 \neq 0 \end{cases}$$

$$x \geq 2 \quad x \neq 1$$

$$x \geq 2 \quad x \neq 1$$

0

8 4 2010 11

1370536875

$$\frac{1.37 \times 10^9}{}$$

1I —

$$a \times 10^n \quad 1 \leq |a| < 10 \quad n \quad a$$

$$n \quad 1 \quad n \quad 1 \quad n \quad a$$

$$1370536875 \quad 1.370536875 \times 10^9 \approx 1.37 \times 10^9$$

$$1.37 \times 10^9$$

$$a \times 10^n \quad 1 \leq |a| < 10 \quad n$$

$$a \quad n$$

$$9 \quad 4 \quad \frac{2}{x-1} = 1 \quad \frac{x-3}{}$$

B3

$$x-1$$

$$x-1$$

$$2 \quad x-1$$

$$x-3$$

$$x-3 \quad x-1 \quad 2 \neq 0$$

$$\therefore x-3$$

1 “ ”

2

$$10 \quad 4 \quad x^2 - 2x - 1 = (x-1+\sqrt{2})(x-1-\sqrt{2})$$

58

11

$$x^2 - 2x - 1 = 0 \quad x = 1 \pm \sqrt{2}$$

$$x^2 - 2x - 1 = 0$$

$$x = 1 \pm \sqrt{2}$$

$$\therefore x^2 - 2x - 1 = [x - 1 + \sqrt{2}][x - 1 - \sqrt{2}] = (x - 1 + \sqrt{2})(x - 1 - \sqrt{2})$$

$$x - 1 - \sqrt{2} \quad x - 1 + \sqrt{2}$$

$$11 \quad 4 \quad x^2 - 4x + a = 0 \quad a = \frac{4}{}$$

AA

$$\Delta=b^2-4ac=0 \qquad a \qquad a$$

$$\Delta=0$$

$$4-4\times 1\times a=0$$

$$a=4$$

$$4$$

$$\Delta=b^2-4ac \qquad \qquad \qquad \Delta$$

$$1 \triangle 0 \Leftrightarrow$$

$$2 \triangle 0 \Leftrightarrow$$

$$3 \triangle 0 \Leftrightarrow$$

$$12-4 \qquad y=\frac{m-3}{x} \qquad x=0 \qquad y=x \qquad m \qquad \underline{m-3}$$

G4

$$y=\frac{m-3}{x} \qquad x=0 \qquad y=x \qquad m-3=0$$

$$y=\frac{m-3}{x} \qquad x=0 \qquad y=x$$

$$\therefore m-3=0$$

$$\therefore m=3$$

$$m=3$$

$$13-4 \qquad \qquad \qquad \text{“} \qquad \text{”} \qquad \qquad \qquad \text{“} \qquad \text{”} \qquad \qquad \qquad x \qquad 60 \leq x < 100$$

$60 \leq x < 70$	30	0.15
$70 \leq x < 80$	m	0.45
$80 \leq x < 90$	60	n
$90 \leq x < 100$	20	0.1

$$n=\underline{0.3}$$

V7

$$11 \quad 27$$

$$60 \leq x < 70 \quad 30 \quad 0.15 \quad n \frac{60}{\text{总人数}}$$

$$\therefore 60 \leq x < 70 \quad 30 \quad 0.15$$

$$\therefore 30 \div 0.15 = 200$$

$$\therefore n \frac{60}{200} = 0.3$$

$$0.3$$

$$\frac{\text{频数}}{\text{总数}}$$

$$14 \quad 4 \quad 6 \quad 120 \quad 180$$

$$x / \quad -\frac{120}{x} - \frac{180}{x+6} -$$

$$B6$$

$$x / \quad 6 \quad 120 \quad 180$$

$$x /$$

$$\frac{120}{x} \quad \frac{180}{x+6}$$

$$\frac{120}{x} \quad \frac{180}{x+6}$$

$$15 \quad 4 \quad ABCD \quad AB \parallel CD \quad E \quad F \quad AD \quad BC \quad \overrightarrow{AB} = \vec{a} \quad \overrightarrow{CD} = \vec{b} \quad \vec{a} \quad \vec{b}$$

$$\overrightarrow{EF} = \frac{1}{2}(\vec{a} + \vec{b})$$

$$LM \quad *$$

$$ABCD \quad AB \parallel CD \quad E \quad F \quad AD \quad BC \quad EF \quad ABCD$$

$$EF \parallel AB \parallel CD \quad EF = \frac{1}{2}(AB + CD) \quad \overrightarrow{AB} = \vec{a} \quad \overrightarrow{CD} = \vec{b}$$

$$\therefore ABCD \quad AB \parallel CD \quad E \quad F \quad AD \quad BC$$

$$\therefore EF \quad ABCD$$

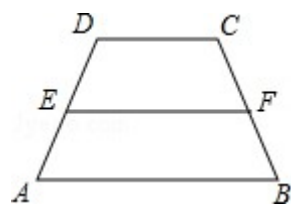
$$\therefore EF \parallel AB \parallel CD \quad EF = \frac{1}{2} (AB + CD)$$

$$\because \overrightarrow{AB} = \vec{a} \quad \overrightarrow{CD} = \vec{b}$$

$$\therefore \overrightarrow{DC} = \vec{b}$$

$$\therefore \overrightarrow{EF} = \frac{1}{2} (\vec{a} + \vec{b})$$

$$\frac{1}{2} (\vec{a} + \vec{b})$$



$$16 - 4 = R - r \quad x^2 - 5x + 6 = 0 \quad 1 \quad \underline{\hspace{1cm}}$$

A8

MJ

$$R - r$$

P

$$R - r \quad R \geq r$$

$$P - R + r$$

$$P - R + r$$

$$R - r \quad P - R + r$$

$$P - R - r$$

$$P - R - r$$

$$\therefore x^2 - 5x + 6 = 0$$

$$\therefore R + r = 5 \quad Rr = 6$$

$$\therefore |R - r| = \sqrt{(R - r)^2} = \sqrt{(R + r)^2 - 4Rr} = \sqrt{5^2 - 4 \times 6} = 1$$

$$\therefore \odot O_1 \cap \odot O_2$$

$$17 - 4 = \triangle ABC \quad G \triangle ABC \quad G \quad DE \parallel BC \quad AB \quad D \quad AC \quad E$$

$$\triangle ABC \quad 9 \quad \triangle ADE \quad \underline{4}$$

K5

S9

$$DE \parallel BC \quad \triangle ADE \triangle ABC$$

2

$$\therefore DE \parallel BC$$

$$\therefore \triangle ADE \sim \triangle ABC$$

$$\therefore G \triangle ABC$$

$$\therefore AG = 2GF$$

$$\therefore AG = \frac{2}{3}AF$$

$$\therefore \frac{AG}{AF} = \frac{2}{3}$$

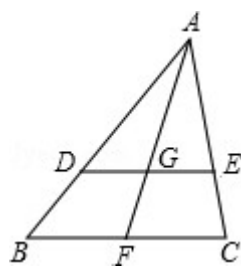
$$\frac{\triangle ADE}{\triangle ABC} = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$$

$$\frac{\triangle ADE \text{ 的面积 }}{\triangle ABC \text{ 的面积 }} = \frac{4}{9}$$

$$\therefore \triangle ABC \text{ 的面积 } = 9$$

$$\therefore \triangle ADE \text{ 的面积 } = \frac{4}{9} \times 9 = 4$$

4



2

$$18 - 4 = 14 \quad \triangle ABC \quad AD = 4 \quad CD = 2 \quad AD = 4 \quad A \quad D \quad CB \quad P \quad \angle DPC$$

$$\frac{75^\circ - 15^\circ}{2}$$

$$KO = 30$$

$$LB$$

$$R2$$

$$A$$

$$D$$

$$CB$$

$$P$$

$$P$$

$$BC$$

$$P$$

$$CB$$

$$P \quad CB$$

$$\therefore AP = AD = 4 \quad AB = CD = 2$$

$$\therefore \triangle ABP \quad \angle APB = 30^\circ$$

$$\therefore AD \parallel BC$$

$$\therefore \angle DAP = \angle APB = 30^\circ$$

∴AP AD

$$\therefore \angle APD = \angle ADP = \frac{180^\circ - 30^\circ}{2} = 75^\circ$$

$$\therefore \angle DPC = 180^\circ - \angle APB - \angle APD = 180^\circ - 30^\circ - 75^\circ = 75^\circ$$

P' CB

$$\triangle AP'B \quad \angle AP'B = 30^\circ \quad \angle P'AB = 60^\circ$$

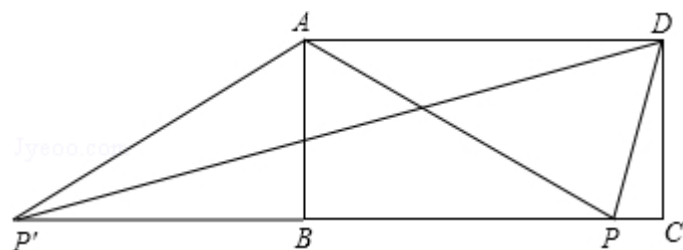
$$\angle P'AD = \angle P'AB + \angle BAD = 60^\circ + 90^\circ = 150^\circ$$

∴AP' AD

$$\therefore \angle AP'D = \angle ADP' = \frac{180^\circ - 150^\circ}{2} = 15^\circ$$

$$\therefore \angle DP'C = \angle AP'B - \angle AP'D = 30^\circ - 15^\circ = 15^\circ$$

$$\angle DPC = 75^\circ - 15^\circ = 60^\circ$$



□□□□□□□□□□ 7 □□□□ 78 □□

$$19 \quad 10 \quad \left(-\frac{1}{4}\right)^{-1} + \sqrt{27} + |\sqrt{3}-3| + \cot 30^\circ$$

2C

6F

T5

4

$$4 + 3\sqrt{3} + 3\sqrt{3} + \sqrt{3}$$

$$3\sqrt{3} - 1$$

$$20 \quad 10 \quad \begin{cases} 3x-7 < 2(1-3x) \\ \frac{x-3}{2} + 1 \leq \frac{3x-1}{4} \end{cases}$$

C2

C4

C6

CB

11

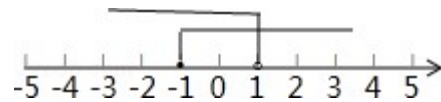
13 23

$$\begin{cases} 3x-7 < 2(1-3x) \text{ ①} \\ \frac{x-3}{2}+1 \leq \frac{3x-1}{4} \text{ ②} \end{cases}$$

$$\because \text{① } x < 1$$

$$\text{② } x \geq 1$$

$$\therefore 1 \leq x < 1$$



21 10

87m

$AD \parallel BC$

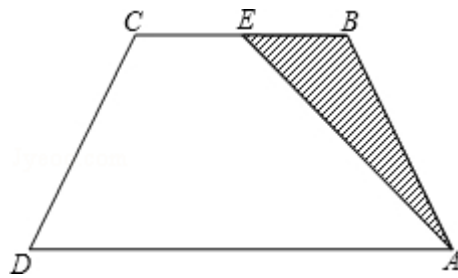
10m AB $i = \frac{5}{3}$

AB

AE $i = \frac{5}{6}$

1 AB

2 EC 2.7m AD



LJ T9

16

1 B $BF \perp AD$ F ABF AF AB

2 E $EG \perp AD$ G EC M AD N MN $S_{\triangle ABE}$ S_{CMND} DN

1 B $BF \perp AD$ F

$$\text{Rt}\triangle ABF \because i = \frac{BF}{AF} = \frac{5}{3} \quad BF = 10m$$

$$\therefore AF = 6m \quad AB = 2\sqrt{34}m$$

$$AB=2\sqrt{34}m$$

$$2 \quad E \quad EG \perp AD \quad G$$

$$\text{Rt}\triangle AEG \because \frac{EG}{AG} = \frac{5}{6} \quad EG=BF=10m$$

$$\therefore AG=12m$$

$$\therefore AF=6m$$

$$\therefore BE=GF=AG-AF=6m$$

$$EC=MC \quad AD=ND \quad MN$$

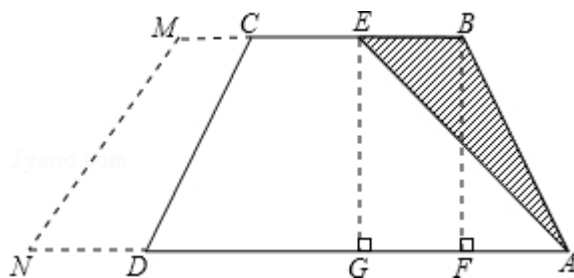
$$\therefore S_{\triangle ABE}=S_{CMND}$$

$$\therefore \frac{1}{2} \cdot BE \cdot EG = \frac{1}{2} (MC+ND) \cdot EG$$

$$BE=MC+ND$$

$$DN=BE-MC=6-2.7=3.3 \text{ m}$$

$$AD=3.3m$$



$$1 \quad B \quad BF \perp AD \quad F \quad \triangle ABF \quad AF=AB=2$$

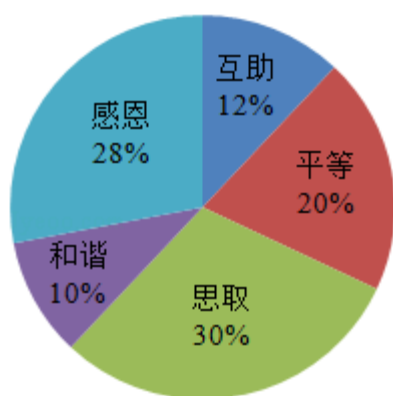
$$S_{\triangle ABE}=S_{CMND} \quad ND$$

$$22-10=12 \quad \text{“ ”} \quad 2 \quad 5 \quad 50$$

$$1 \quad \text{“ ”} \quad \underline{5} \quad \text{“ ”} \quad \underline{36^\circ}$$

$$2 \quad 1500 \quad \text{“ ”} \quad \underline{420}$$

$$3 \quad 5 \quad \text{“ ” “ ”}$$



V5 VB X6
 27
 1 “ ”
 2
 3
 1 50 “ ” 10%
 50×10% 5 360°×10% 36°

2∴“ ” 28%
 ∴1500×28% 420

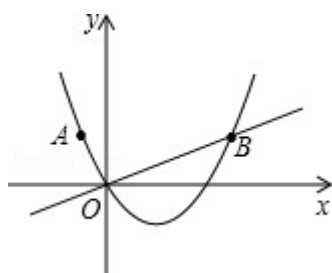
3

∴“ ” “ ” $\frac{1}{10}$

23 12 A 1 2 O $l\perp OA$ A O B
 OB 2OA

1

2 $BC \perp l$ C x m m 0 l P Q B C P Q
 m



HF

15

1 A $AH \perp l$ H B $BC \perp l$ C A AH OH
 $\triangle AHO \sim \triangle OCB$ $OB = 2OA$ B
 2 l B C P Q PQ BC m

1 A $AH \perp l$ H B $BC \perp l$ C

A 1 2 AH 2 OH 1

$OB \perp OA$ $\triangle AHO \sim \triangle OCB$

$$\frac{AH}{OC} = \frac{OH}{BC} = \frac{OA}{OB}$$

$\because OB = 2OA$

$\therefore OC = 4$ $BC = 2$

$\therefore B = 4$ 2

A O B $y = ax^2 + bx + c$ $a \neq 0$

$$\begin{cases} a - b + c = 2 \\ 16a + 4b + c = 2 \\ c = 0 \end{cases}$$

$$\begin{cases} a = \frac{1}{2} \\ b = -\frac{3}{2} \\ c = 0 \end{cases}$$

$$y = \frac{1}{2}x^2 - \frac{3}{2}x$$

$$2 \quad l \quad y = kx \quad k \neq 0$$

$$\therefore l \quad B(4, 2)$$

$$\therefore l \quad y = -\frac{1}{2}x$$

$$\therefore x = m \quad m \neq 0 \quad l \quad P \quad Q$$

$$\therefore P \quad (m, \frac{1}{2}m) \quad Q \quad (m, \frac{1}{2}m^2 - \frac{3}{2}m)$$

$$\therefore B, C, P, Q$$

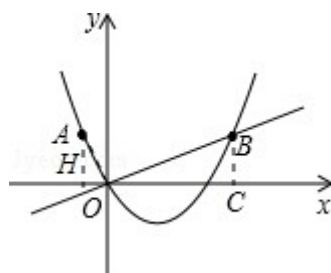
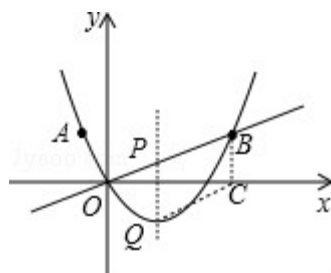
$$\therefore PQ \parallel BC \quad PQ \perp BC$$

$$|\frac{1}{2}m - (\frac{1}{2}m^2 - \frac{3}{2}m)| = 2$$

$$m = 2 \pm 2\sqrt{2} \quad m \neq 2$$

$$\therefore m \neq 0$$

$$\therefore m = 2\sqrt{2} + 2 \quad 2$$



$$24 \quad 12 \quad \triangle ABC \quad \angle ABC = 90^\circ \quad E \quad AC$$

$$C \quad BE$$

$$A \quad BE$$

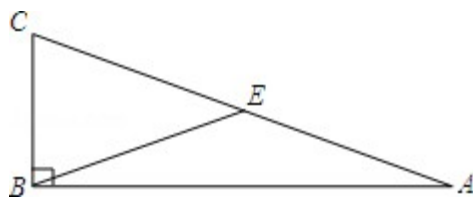
$$D \quad AD$$

$$DF \quad BE$$

$$EF \quad BD$$

$$1 \quad EF \quad BD$$

$$2 \quad AF \quad 13 \quad CD \quad 6 \quad AC$$



AD KP KQ L6 LA

11

1 $BEDF$ $BE \parallel DF$ $BEDF$

2 $DF = BE = x$ $AC = 2x$ $AD = AF = DF = 13 - x$ $\text{Rt}\triangle ACD$ x

1 $EF \parallel BD$

$DE = BF \because BE \parallel DF$ $BE = DF$

$\therefore BEDF$

$\because CD \perp BE$

$\therefore CD \perp AD$

$\because \angle ABC = 90^\circ$ $E \in AC$

$\therefore BE = DE = \frac{1}{2}AC$

$\therefore BEDF$

$\therefore EF \parallel BD$

2 $DF = BE = x$ $AC = 2x$ $AD = AF = DF = 13 - x$

$\text{Rt}\triangle ACD \because AD^2 + CD^2 = AC^2$

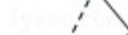
$\therefore (13 - x)^2 + 6^2 = (2x)^2$

$3x^2 + 26x - 205 = 0$

$x_1 = \frac{41}{3}$ $x_2 = 5$

$\therefore AC = 10$

$AC = 10$



$$\therefore \widehat{AC} = \widehat{CD}$$

$$\therefore \angle P = \angle POC$$

$$\therefore \angle CPO = \angle COD$$

$$\angle PDO = \angle ODC$$

$$\therefore \triangle DOC \sim \triangle DPO$$

$$\therefore DC \cdot OD = OD \cdot DP \quad OD^2 = DC \cdot DP$$

$$\therefore DC \cdot DC + 4 = 16$$

$$\therefore CD = 2\sqrt{5} - 2$$

$$2 \quad OE \perp CD \quad E \quad 1$$

$$CE = \frac{1}{2}CD = \frac{1}{2}y$$

$$\therefore \angle P = \angle P = \angle PBC = \angle PEO = 90^\circ$$

$$\therefore \triangle PBC \sim \triangle PEO$$

$$\therefore \frac{PB}{PE} = \frac{PC}{OP}$$

$$PB = \frac{1}{2}OP = \frac{1}{2}(x+4) \quad PE = PC + CE = 4 + \frac{1}{2}y$$

$$\therefore \frac{\frac{x+4}{2}}{4 + \frac{y}{2}} = \frac{4}{x+4}$$

$$\therefore y = \frac{1}{4}x^2 + 2x - 4 = 4\sqrt{2} - 4 \quad x = 4$$

$$3 \quad D \quad \widehat{AC}$$

$$OC = OE$$

$$Rt\triangle CBP \cong Rt\triangle CBO$$

$$\therefore \angle CPB = \angle COB = x \quad x$$

$$\therefore \angle DCO = 2x$$

$$PC = OC = R = 4$$

$$\therefore CE = DE$$

$$\therefore OE \perp CD$$

$$\triangle Rt\triangle OEC \cong Rt\triangle OED$$

$$\therefore \begin{cases} EO=EO \\ CO=DO \\ CE=ED \end{cases}$$

$$\therefore \text{Rt}\triangle OEC \cong \text{Rt}\triangle OED \quad \text{SSS}$$

$$\therefore \angle ODC = \angle OCD = 2x$$

$$\text{Rt}\triangle OPE$$

$$\tan \angle APD = \frac{OE}{PE}$$

$$\therefore \angle CBO = \angle CEO = 90^\circ$$

$$\therefore B, C, E, O$$

$$\therefore$$

$$\angle BEC = \angle BOC = x$$

$$\therefore \angle DEF = \angle BEC = \angle BOC = x$$

$$\triangle DEF$$

$$\angle ODC = \angle F + \angle DEF$$

$$\therefore 2x = \angle F + x$$

$$\therefore \angle F = x$$

$$\therefore \triangle DEF$$

$$CE = DE = DF = 1$$

$$\therefore PE = PC + CE = 4 + 1 = 5$$

$$\text{Rt}\triangle ODE \quad DE = 1 \quad OD = R = 4$$

$$\therefore OE = \sqrt{15}$$

$$\therefore \tan \angle P = \frac{OE}{PE} = \frac{\sqrt{15}}{5}$$

$$D \quad \widehat{AC}$$

$$CE = DE = DF = 1 \quad PC = OC = r = 4$$

$$PE = 3 \quad OE = \sqrt{15}$$

$$\tan \angle P = \frac{OE}{PE} = \frac{\sqrt{15}}{3}$$

$$\tan \angle P = \frac{\sqrt{15}}{5} \cdot \frac{\sqrt{15}}{3}$$

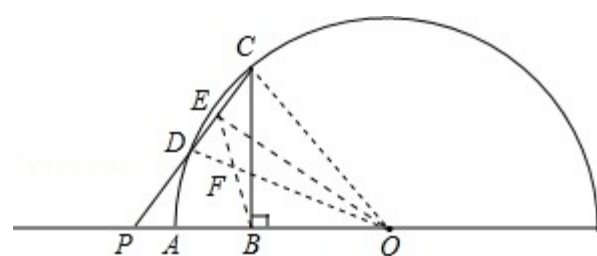


图3

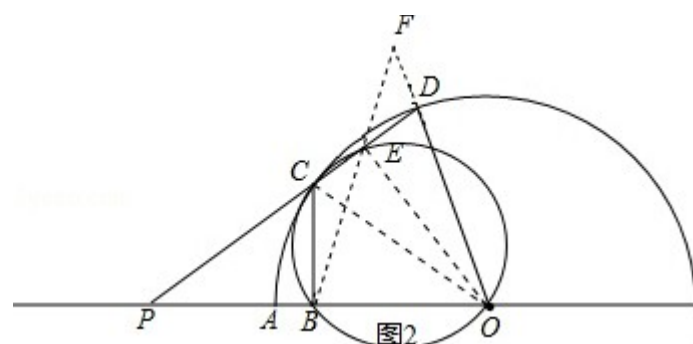


图2

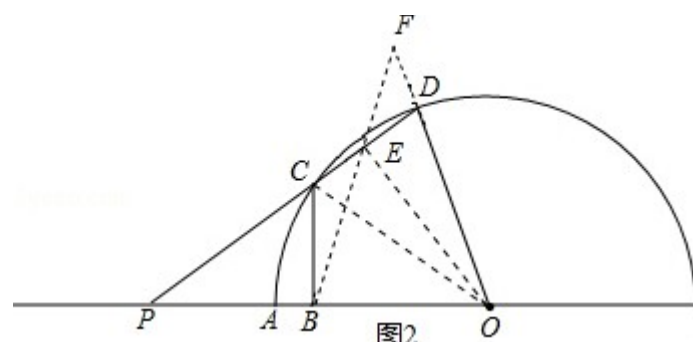


图2

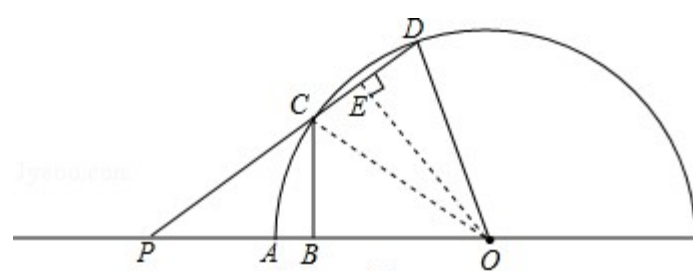


图1